A Symbolic–Affine Framework for Density-One Descent in the Accelerated Collatz Map

Wayne Brassem

August 5, 2026

Abstract

We study the Collatz map through the fully accelerated transformation

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

acting on the odd positive integers.

Finite accelerated trajectories are encoded by *division words*, which record the sequence of 2-adic valuations encountered along an orbit. Each admissible division word determines a canonical affine transformation together with a unique congruence class modulo a power of two, yielding a symbolic–affine representation of accelerated Collatz dynamics.

The admissible division words form a constrained symbolic language whose admissibility is governed by the Beatty threshold sequence associated with $\log_2 3$. This Beatty grammar admits a canonical Sturmian return-word decomposition determined by the continued-fraction expansion of $\log_2 3$. The resulting return blocks yield a spectral bound

$$\lambda < 3$$

for the exponential growth rate of admissible symbolic configurations. Comparing this spectral bound with the exponential dyadic refinement of the associated congruence cylinders shows that symbolic proliferation is strictly dominated by dyadic contraction

Consequently, the set of positive integers admitting an eventual descent under the accelerated Collatz map has natural density one. The argument does not exclude the existence of exceptional trajectories, but shows that any such exceptions must belong to a residual set of natural density zero.

Contents

1 Introduction

The $3x + 1$ problem, also known as the Collatz conjecture [1], concerns the behavior of the map

$$T(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ 3x + 1, & \text{if } x \text{ is odd.} \end{cases}$$

on the positive integers. It asserts that every positive integer eventually reaches 1 under iteration of T. Despite its elementary formulation, the conjecture has resisted proof for decades [2, 3]. Extensive computational verification confirms convergence for very large ranges [4], but does not reveal the global structure governing trajectory behavior.

Recently, a major advance by Tao [5] established that almost all integers eventually attain values smaller than their initial value under iteration of the Collatz map in the sense of logarithmic density.

The present work develops a deterministic, arithmetic framework for analyzing typical Collatz behavior based on a symbolic–affine encoding of trajectories. Rather than studying individual trajectories directly, we encode the fully accelerated dynamics symbolically using the map

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

which maps odd integers to odd integers by compressing each odd step into a single transformation.

Each trajectory under f is represented by a finite *division word*, recording the number of factors of 2 removed (the 2-adic valuation) at each iterate. These words admit an exact affine representation obtained by symbolic composition of the accelerated map: each division word, ω, induces a map of the form

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

and determines a corresponding congruence class modulo $2^{D(\omega)}$. Thus division words determine canonical arithmetic *residue classes* in the integers, once realizability is imposed.

Division words are *admissible* if they arise from actual trajectories under the accelerated map. Admissibility is governed by the Beatty [6] threshold sequence

$$B(n) = \lfloor n \log_2 3 \rfloor + 1$$

whose increment sequence is a Sturmian [7] word. Consequently the admissible symbolic grammar inherits the combinatorial constraints of an irrational rotation rather than arising from an ad hoc combinatorial construction. This connection places the admissibility problem within the well-developed theory of Beatty sequences and Sturmian symbolic dynamics.

Congruence classes allow the dynamics to be studied at the level of residue classes rather than individual orbits. Convergent division words—those for which the associated affine map is contractive—identify families of integers that decrease after a fixed number of steps. The global problem is thereby reduced to understanding how these families are distributed and how their associated densities accumulate.

We then show that the symbolic growth generated by the admissible grammar is strictly subcritical: the number of admissible division words grows like λ^m with $\lambda < 3$, while the corresponding congruence cylinders undergo exponential dyadic refinement. The central mechanism of the paper is the competition between these two exponential processes.

Admissible division words organize naturally into a forest of nested cylinders. Contractive words define first-passage cylinders that remove positive measure from the residual set, while non-contractive extensions form a shrinking residual forest.

This imbalance forces the residual forest to become exponentially sparse. Symbolic branching cannot compensate for the increasingly fine arithmetic subdivision imposed by successive congruence constraints.

The central result of the paper is that the admissible Beatty grammar has spectral growth strictly below the rate of dyadic refinement. This reduces the density question to a comparison of two exponential processes and implies that the residual forest has natural density zero. Consequently, the set of integers admitting eventual descent under the accelerated Collatz map has natural density one.

While this does not exclude the existence of exceptional trajectories, any such exceptions must lie within a set of zero density and cannot arise from any scalable arithmetic structure generated by the symbolic dynamics. Any counterexample to the Collatz Conjecture, if it exists, must therefore avoid every contractive cylinder generated by the affine grammar and remain confined to the residual forest.

Structure of the Paper. The argument proceeds as follows:

- Section 2 introduces division words and their total division counts.

- Section 3 develops the affine representation of division words and their congruence structure.
- Section 4 establishes the stepwise realizability of admissible division words and the unique residue lifting property.
- Section 5 defines the geometry of contractive cylinders.
- Section 6 discusses competition between symbolic entropy and dyadic growth.
- Section 7 establishes the collapse of measure for disjoint cylinders.
- Section 8 develops the Beatty grammar governing admissible symbolic growth and establishes the Beatty Barrier Theorem.
- Section 9 proves the spectral growth bound and combines it with the measure estimates to establish collapse of the residual forest.

2 Symbolic Encoding of Collatz Trajectories

2.1 Division Words

Definition 2.1 (Division Word)**.** A division word of length $m \geq 1$ is a sequence

$$\omega = (d_0, \ldots, d_{m-1}),$$

where each $d_i \in \mathbb{N}$ is the 2-adic valuation

$$v_2(3f^i(x) + 1)$$

of the corresponding step in the accelerated Collatz map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Remark. Since f maps odd integers to odd integers, each $3f^i(x)+1$ is even, so $d_i \geq 1$ for all i.

Definition 2.2 (Realization)**.** An odd integer $x \in \mathbb{N}$ realizes the division word ω if

$$d_i = v_2(3f^i(x) + 1), \quad 0 \leq i < m.$$

Remark. Every odd integer x induces a unique infinite valuation sequence

$$(v_2(3x+1),\, v_2(3f(x)+1),\, v_2(3f^2(x)+1), \ldots),$$

of which a division word is a finite prefix.

2.2 Word Length and Total Division Count

Definition 2.3 (Length)**.** The length of a division word

$$\omega = (d_0, \dots, d_{m-1})$$

is

$$|\omega| := m.$$

Definition 2.4. The total division count of a word ω is

$$D(\omega) := \sum_{i=0}^{m-1} d_i.$$

Remark. The two basic numerical invariants of a division word are its length $|\omega|$, which records the number of accelerated Collatz steps, and its total division count $D(\omega)$, which records the cumulative dyadic exponent. The interaction between these two quantities governs both the affine dynamics and the symbolic geometry developed below.

3 Affine Representation of Division Words

Definition 3.1 (Affine Map)**.** Each (not necessarily realizable) division word ω of length m determines a formal affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

obtained by symbolic composition of the accelerated Collatz map.

Remark. The quantity $D(\omega)$ records the total exponent of 2 removed along the trajectory segment encoded by ω. It determines the modulus $2^{D(\omega)}$ appearing in the associated affine representation above.

A division word is called *realizable* if it is realized by at least one odd integer. In later sections we use the synonymous term *admissible* when emphasizing the arithmetic structure of realizable division words.

3.1 Concatenation and Prefix Structure

Proposition 3.1 (Concatenation and Nested Affine Structure)**.** Let ω and η be division words. If the concatenation $\omega\eta$ denotes the word obtained by appending η to ω, then the associated affine maps satisfy

$$F_{\omega\eta} = F_\eta \circ F_\omega.$$

Consequently, concatenation of division words is compatible with composition of their affine maps: the affine map associated with a concatenated word is exactly the composition of the affine maps associated with its constituent words.

Furthermore, if C_ω denotes the set of integers realizing the word ω, then

$$C_{\omega\eta} \subseteq C_\omega.$$

Thus extensions of division words correspond to nested realization sets.

Proof. Write

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

and similarly

$$F_\eta(x) = \frac{3^n x + c(\eta)}{2^{D(\eta)}}.$$

Then

$$F_\eta(F_\omega(x)) = \frac{3^n \left(\frac{3^m x + c(\omega)}{2^{D(\omega)}}\right) + c(\eta)}{2^{D(\eta)}}.$$

Rearranging gives

$$F_\eta(F_\omega(x)) = \frac{3^{m+n} x + 3^n c(\omega) + 2^{D(\omega)} c(\eta)}{2^{D(\omega)+D(\eta)}}.$$

The denominator and ternary exponents add under concatenation, while the affine constant combines according to the usual affine composition rule. Therefore this map is precisely the affine representation associated with the concatenated word $\omega\eta$.

For the inclusion of realization sets, any integer realizing $\omega\eta$ must first realize the prefix ω before realizing the appended word η. Hence every realization of $\omega\eta$ is also a realization of ω, and therefore

$$C_{\omega\eta} \subseteq C_\omega.$$

$\square$

Definition 3.2 (Contractive and Expansive Affine Maps)**.** Let ω be a realizable division word of length m. Since

$$F_\omega(x) = f^m(x)$$

for every realization of ω, the affine map F_ω is called

- contractive if $F_\omega(x) < x$ for every realization of ω;
- expansive if $F_\omega(x) \geq x$ for every realization of ω.

Remark. Contractivity refers only to the finite trajectory encoded by ω. Thus

$$F_\omega(x) = f^m(x) < x,$$

but no conclusion is drawn about the behavior of later iterates or eventual convergence to 1.

3.2 Explicit Form of the Affine Constant

Remark (Division Words as a Dynamical Encoding)*.* Recall from Section 2 that a *division word* $\omega = (d_0, \ldots, d_{m-1})$ records the exact 2-adic valuation $d_i = v_2(3x_i + 1)$, occurring at each iterate of the fully accelerated Collatz map.

Division words retain the dynamical information needed to reconstruct the affine representation. Each division word $\omega = (d_0, \ldots, d_{m-1})$ determines the sequence of multiplicative factors of the linear coefficient

$$r_i = \frac{3}{2^{d_i}},$$

so that the multiplicative growth or decay contribution of each iterate is preserved in the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure allows symbolic trajectories to be analyzed through their affine representations and forms the foundation for the realizability and descent results developed later.

Lemma 3.1 (Explicit Form of the Affine Constant)**.** *Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word. Then the affine map*

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

has the explicit form

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i},$$

where empty sums are defined as 0*.*

Proof. Iterating the accelerated map symbolically gives

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}.$$

Repeated substitution yields

$$x_m = \frac{3^m x_0}{2^{D(\omega)}} + \sum_{j=0}^{m-1} \frac{3^{m-1-j}}{2^{\sum_{i=j}^{m-1} d_i}}.$$

Multiplying by $2^{D(\omega)}$ gives

$$2^{D(\omega)} x_m = 3^m x_0 + \sum_{j=0}^{m-1} 3^{m-1-j} 2^{\sum_{i=0}^{j-1} d_i},$$

which identifies the affine constant $c(\omega)$. $\square$

Remark. The affine constant $c(\omega)$ records the accumulated contribution of every additive term introduced during symbolic iteration. Earlier contributions are multiplied by more subsequent factors of 3, whereas later contributions inherit larger powers of 2 from the preceding divisions.

3.3 Dyadic-Ternary Balance

The symbolic dynamics of a division word are governed by the balance between the cumulative ternary growth 3^m and the cumulative dyadic contraction $2^{D(\omega)}$. For a division word ω of length m, the associated affine map has linear coefficient

$$\frac{3^m}{2^{D(\omega)}}.$$

The linear coefficient determines the asymptotic behavior of the affine map:

- if $2^{D(\omega)} < 3^m$: expansive slope.
- if $2^{D(\omega)} > 3^m$: contractive slope.

This naturally introduces the Beatty threshold sequence

$$B(m) = \lfloor m \log_2 3 \rfloor + 1,$$

which is the OEIS sequence A020914. This is the minimal integer k satisfying

$$2^k > 3^m.$$

Consequently, $B(m) = A020914(m)$ represents the minimal total division count for which the affine slope becomes strictly contractive.

3.4 Example

Consider convergent segments with $m = 4$ iterates of f. The minimal total division count for $m = 4$ is

$$A020914(4) = 7.$$

One realizable division word attaining this minimum is

$$\omega = (1, 1, 1, 4),$$

for which

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Composing the accelerated map according to ω yields

$$f_\omega(x) = \frac{3^4 x + c(\omega)}{2^7} = \frac{81x + 65}{128},$$

The integrality condition

$$81x + 65 \equiv 0 \pmod{128}$$

determines a unique residue class

$$x \equiv 15 \pmod{128}.$$

This congruence determines a unique residue class modulo 2^7, and defines the arithmetic constraint associated with the division word. In Section 5, these congruence classes will be identified with the cylinder representation of admissible words.

Interpretation

The word $\omega = (1, 1, 1, 4)$ records three successive divisions by 2, followed by a division by 2^4. The affine representation packages this symbolic information into a single arithmetic transformation together with its associated congruence condition.

Thus $d_3 = 4$ means that the fourth iterate removes exactly four factors of 2. The constant $c(\omega)$ is fixed entirely by this symbolic structure and ensures that precisely the integers in the residue class $x \equiv 15 \pmod{128}$ realize the affine transformation associated with ω.

This example illustrates that the behavior of trajectories is governed entirely by the symbolic data encoded in ω and its associated affine map. The initial values that realize a given word are precisely those satisfying the induced congruence constraint, independent of any particular choice of representative.

The affine representation converts symbolic trajectory data into explicit arithmetic constraints modulo powers of two, allowing realizable trajectories to be studied through congruence geometry rather than direct iteration.

The affine representation is purely symbolic: it associates an affine transformation with every formal division word. The following sections determine which formal words are realizable under the accelerated Collatz map by developing the admissible division grammar and proving its equivalence to realizability.

4 Stepwise Realizability and Admissibility

4.1 Stepwise Realizability

Theorem 4.1 (Affine Stepwise Identity). *Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word, and define $c(\omega)$ by*

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i}.$$

Suppose

$$v_2(3x+1) = d_i, \qquad 0 \le i < m.$$

Equivalently,

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \qquad 0 \le i < m.$$

Then

$$2^{D(\omega)} x_m = 3^m x_0 + c(\omega).$$

Corollary 4.2 (Stepwise Integrality Criterion). *The sequence $(x_0, \ldots, x_m)$ is an integer trajectory under the accelerated Collatz relations if and only if*

$$3^m x_0 + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Remark (Structural Interpretation). Each term in $c(\omega)$ records the propagated contribution of a single additive "+1" created during iteration. Earlier contributions accumulate additional powers of 3 through forward propagation and additional powers of 2 through delayed subsequent divisions by powers of two, producing a fully time-ordered encoding of the trajectory in a single affine constant.

4.2 Division Word Valuation

Lemma 4.3 (Exact Valuation Congruence)**.** *Let x be an odd integer and let $d \geq 1$. Then*

$$v_2(3x+1) = d$$

if and only if

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Equivalently,

$$3x+1 = 2^d(2k+1)$$

for some integer k.

Proof. Suppose first that

$$v_2(3x+1) = d.$$

Then by definition,

$$3x+1 = 2^d y,$$

where y is odd. Writing

$$y = 2k+1,$$

gives

$$3x+1 = 2^d(2k+1) = 2^{d+1}k + 2^d,$$

and therefore

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Conversely, suppose that

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Then there exists an integer k such that

$$3x+1 = 2^{d+1}k + 2^d = 2^d(2k+1).$$

Since $2k + 1$ is odd, exactly d factors of 2 divide $3x + 1$. Hence

$$v_2(3x + 1) = d.$$

□

Corollary 4.4 (Local Valuation Refinement)**.** *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word and let

$$\omega_j = (d_0, \dots, d_{j-1})$$

denote its prefixes. For each j, the condition

$$v_2(3f^j(x) + 1) = d_j$$

is equivalent to the congruence

$$3f^j(x) + 1 \equiv 2^{d_j} \pmod{2^{d_j+1}}.$$

Consequently, each additional valuation constraint refines the realization set of the prefix ω_j to a smaller realization set associated with the extended prefix

$$\omega_{j+1}.$$

Thus admissible division words generate a nested sequence of arithmetic constraints, and extensions of division words correspond to successive refinements of residue classes.

Remark. The corollary formalizes the local nature of admissibility. Exact valuation constraints are imposed one step at a time and depend only on the current accelerated iterate. Consequently, realizability propagates forward recursively through the valuation sequence rather than being determined solely by a terminal congruence condition.

Theorem 4.5 (Residue Lifting Theorem)**.** *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word. Then there exists a unique residue class

$$r_\omega \pmod{2^{D(\omega)}}$$

consisting precisely of the integers realizing ω. Moreover, if

$$\omega_j = (d_0, \dots, d_{j-1})$$

denotes the prefix of length j, then the corresponding residue classes satisfy

$$r_{\omega_{j+1}} \equiv r_{\omega_j} \pmod{2^{D(\omega_j)}}.$$

Thus each additional valuation constraint uniquely refines the residue class of its prefix.

Proof. We proceed by induction on the length m of the division word. For $m = 1$, the word consists of a single valuation

$$\omega = (d_0).$$

By Lemma 4.3,

$$v_2(3x + 1) = d_0$$

if and only if

$$3x + 1 \equiv 2^{d_0} \pmod{2^{d_0+1}}.$$

Since 3 is invertible modulo every power of two, this congruence has a unique solution modulo 2^{d_0+1}. Thus the realization set of ω is a single residue class.

Assume now that the result holds for all admissible words of length m, and let

$$\omega' = (d_0, \dots, d_m)$$

be an admissible word of length $m + 1$. Let

$$\omega = (d_0, \dots, d_{m-1})$$

be its length-m prefix. By the induction hypothesis, the integers realizing ω form a unique residue class

$$r_\omega \pmod{2^{D(\omega)}}.$$

The additional valuation condition

$$v_2(3f^m(x) + 1) = d_m$$

is equivalent, by Lemma 4.3, to

$$3f^m(x) + 1 \equiv 2^{d_m} \pmod{2^{d_m+1}}.$$

Using the affine representation of the prefix,

$$f^m(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

this becomes a linear congruence condition on x modulo

$$2^{D(\omega)+d_m} = 2^{D(\omega')}.$$

Since 3^m is odd, it is invertible modulo every power of two. This congruence determines a unique refinement of the residue class

$$r_\omega \pmod{2^{D(\omega)}}$$

to a residue class

$$r_{\omega'} \pmod{2^{D(\omega')}}.$$

Therefore every admissible extension uniquely refines the residue class of its prefix. The result follows by induction. □

4.3 Affine Realization

Theorem 4.6 (Affine Realization Theorem)**.** *Let ω be a division word.*

An integer x realizes ω if and only if

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Proof. Assume that

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

By the Exact Valuation Congruence Lemma 4.3, the residue class associated with the full word ω is contained in the residue class of every prefix

$$\omega_j = (d_0, \dots, d_{j-1}).$$

Therefore, for every prefix length j,

$$x \equiv r_{\omega_j} \pmod{2^{D(\omega_j)}}.$$

We prove inductively that the successive iterates generated from x realize the prescribed valuations. For $j = 0$, the prefix condition gives

$$x \equiv r_{(d_0)} \pmod{2^{d_0}},$$

and therefore

$$2^{d_0} \mid 3x + 1.$$

Because the residue class is defined using the exact division count d_0, the next higher power does not divide:

$$2^{d_0+1} \nmid 3x + 1.$$

Hence

$$v_2(3x + 1) = d_0.$$

Therefore the first accelerated iterate is

$$x_1 = \frac{3x_0 + 1}{2^{d_0}},$$

and is an integer. Now assume that the first j steps have been realized, so that

$$x_j = f^j(x)$$

is the integer determined by the prefix

$$(d_0, \ldots, d_{j-1}).$$

The prefix congruence condition for

$$\omega_{j+1} = (d_0, \ldots, d_j)$$

implies that

$$2^{d_j} \mid 3x_j + 1.$$

Again, exactness of the prefix residue condition excludes the higher power:

$$2^{d_j+1} \nmid 3x_j + 1.$$

Consequently,

$$v_2(3x_j + 1) = d_j,$$

and the next iterate

$$x_{j+1} = \frac{3x_j + 1}{2^{d_j}}$$

is an integer. By induction, every step satisfies the prescribed valuation constraint, and therefore

$$v_2(3f^j(x) + 1) = d_j, \qquad 0 \le j < m.$$

Hence x realizes the division word ω. $\square$

4.4 Admissibility of Division Words

Definition 4.1 (Admissible Division Word)**.** A division word is *admissible* if it is realizable.

Remark. Equivalently, a division word ω is admissible if its induced affine residue class contains an integer whose successive accelerated iterates satisfy

$$v_2(3x_i + 1) = d_i, \qquad 0 \le i < m.$$

Thus admissibility is precisely the condition that a formal symbolic word occurs as an actual finite valuation sequence of the accelerated Collatz map.

4.5 Residue Class Representation of Division Words

Lemma 4.7 (Prefix Nesting of Realization Sets)**.** *Let ω, ω' be admissible division words.*

If the realization sets of ω and ω' intersect, then one word is a prefix of the other.

Proof. Every odd integer x determines a unique infinite valuation sequence

$$(v_2(3x + 1),\, v_2(3f(x) + 1),\, v_2(3f^2(x) + 1), \dots).$$

If x lies in the realization sets of both ω and ω', then both words must agree with the initial segment of this valuation sequence.

Therefore the two words cannot diverge at any position before the shorter word terminates. Hence one word is a prefix of the other. $\square$

5 Cylinder Geometry of the Affine Grammar

The admissible division-word grammar is organized by 2-adic residue classes induced by admissible division words.

These residue classes provide an arithmetic realization of the symbolic dynamics and naturally suggest a geometric interpretation in terms of nested congruence classes, developed below.

5.1 Cylinder of a Division Word

Definition 5.1 (Cylinder)**.** Let ω be an admissible division word with associated residue class

$$r_\omega \pmod{2^{D(\omega)}}.$$

The cylinder associated with ω is

$$C_\omega := \{x \in \mathbb{N} : x \equiv r_\omega \pmod{2^{D(\omega)}}\}.$$

5.2 Prefix Structure and Forest Geometry of Cylinders

Lemma 5.1 (Prefix Nesting of Cylinders)**.** *Let ω and ω' be admissible division words.*

If

$$C_\omega \cap C_{\omega'} \neq \varnothing,$$

then one of the words is a prefix of the other.

Proof. Suppose

$$x \in C_\omega \cap C_{\omega'}.$$

Then x realizes both division words. By the Affine Realization Theorem 4.6, C_ω and $C_{\omega'}$ are precisely the realization sets of ω and ω'. The result therefore follows immediately from Lemma 4.7. □

Theorem 5.2 (Cylinder Inclusion Criterion)**.** *Let ω and η be admissible division words.*

Then

$$C_\eta \subseteq C_\omega$$

if and only if

$$\omega$$

is a prefix of

$$\eta.$$

Proof. Suppose first that ω is a prefix of η. Any integer realizing η must realize every prefix of η, including ω. Therefore

$$C_\eta \subseteq C_\omega.$$

Conversely, suppose

$$C_\eta \subseteq C_\omega.$$

Let

$$x \in C_\eta.$$

Then

$$x \in C_\eta \cap C_\omega.$$

By Lemma 5.1, one word is a prefix of the other. If η were a prefix of ω, then applying the forward direction already established would imply

$$C_\omega \subseteq C_\eta.$$

Together with

$$C_\eta \subseteq C_\omega,$$

this would force

$$C_\eta = C_\omega,$$

Since each cylinder is defined by a unique residue class modulo $2^{D(\omega)}$, equality of cylinders implies equality of moduli. Thus

$$D(\eta) = D(\omega).$$

Because one word is a prefix of the other and the two words have equal depth, they must coincide:

$$\eta = \omega.$$

Therefore ω must be a prefix of η. $\square$

Corollary 5.3 (Disjointness of Sibling Cylinders)**.** *If neither admissible word is a prefix of the other, then*

$$C_\omega \cap C_\eta = \varnothing.$$

Proof. If the cylinders intersected, then by Lemma 5.1, one word would be a prefix of the other, contradicting the assumption. □

Corollary 5.4 (Disjointness at Equal Depth)**.** *Distinct admissible division words of equal length define disjoint cylinders.*

Proof. If two equal-length words intersected, then by Lemma 5.1, one would be a prefix of the other. Equal lengths force equality of the words, contradicting distinctness. □

Corollary 5.5 (Symbolic-Arithmetic Correspondence)**.**

$$\omega \preceq \eta \iff C_\eta \subseteq C_\omega.$$

Proof. This is exactly the content of Theorem 5.2. □

Remark. The previous results identify symbolic ancestry with arithmetic containment. Thus admissible division words organize into a collection of nested 2-adic cylinders whose inclusion relations coincide exactly with the prefix order on division words.

The admissible symbolic grammar therefore forms a disjoint *forest* of prefix trees rather than a single global tree. Each node corresponds to an admissible one-step refinement of a cylinder, obtained by extending the associated division word by one admissible symbol. Each descendant cylinder is obtained by refining the associated congruence class modulo a higher power of two.

5.3 Contractive Cylinders

Definition 5.2 (Contractive Cylinder)**.** A cylinder C_ω is called linearly contractive if its associated affine map

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}$$

has contractive linear coefficient:

$$\frac{3^{|\omega|}}{2^{D(\omega)}} < 1.$$

Equivalently,

$$D(\omega) > |\omega| \log_2 3.$$

5.4 Cylinder Contraction Threshold

Lemma 5.6 (Minimal Contraction Threshold)**.** *Let ω be an admissible division word of length m. If the associated affine map*

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

is contractive in its linear coefficient, then

$$D(\omega) \geq A020914(m),$$

where

$$B(m) = A020914(m) = \lfloor m \log_2 3 \rfloor + 1$$

is the minimal integer k satisfying

$$2^k > 3^m.$$

Proof. Contractivity requires

$$\frac{3^m}{2^{D(\omega)}} < 1.$$

which is equivalent to

$$2^{D(\omega)} > 3^m.$$

The minimal integer satisfying this inequality is exactly $A020914(m)$, so

$$D(\omega) \geq B(m) = A020914(m).$$

□

Remark. The Beatty threshold sequence A020914 governs the growth/decay balance along admissible branches within every cylinder of the forest.

6 Symbolic Growth of the Cylinder Forest

The cylinder forest introduced in the previous section exhibits two competing exponential effects:

1. combinatorial growth in the number of admissible cylinders; and
2. dyadic contraction in the size of individual cylinders.

The first of these is purely symbolic and is quantified below.

6.1 Symbolic Complexity

Definition 6.1 (Symbolic complexity)**.** Let $N(m)$ denote the number of admissible division words of length m. Equivalently, $N(m)$ is the number of admissible cylinders appearing at depth m in the symbolic forest.

Remark. The quantity $N(m)$ measures the combinatorial complexity of the admissible grammar at depth m. It counts the number of distinct admissible division words, or equivalently the number of admissible cylinders obtained after m levels of symbolic refinement.

6.2 Symbolic Growth Rate

Definition 6.2 (Symbolic Growth Rate)**.** Define the symbolic growth rate of the admissible grammar by

$$\lambda := \limsup_{m\to\infty} N(m)^{1/m}.$$

Remark. The quantity λ is the exponential branching rate of the admissible cylinder forest. It measures the asymptotic combinatorial complexity of admissible symbolic extensions.

Lemma 6.1 (Uniform exponential envelope)**.** *For every $\varepsilon > 0$, there exists m_0 such that*

$$N(m) \le (\lambda + \varepsilon)^m$$

for all $m \ge m_0$.

Proof. This is an immediate consequence of the definition of

$$\lambda := \limsup_{m\to\infty} N(m)^{1/m}.$$

□

6.3 Entropy Versus Dyadic Contraction

Each admissible cylinder is determined by a congruence class modulo

$$2^{D(\omega)}$$

Consequently, deeper cylinders correspond to congruence classes modulo progressively larger powers of two. Since

$$D(\omega) \geq B(m)$$

every cylinder at depth m is defined modulo at least

$$2^{B(m)}$$

where

$$B(m) = A020914(m) = \lfloor m \log_2 3 \rfloor + 1$$

therefore

$$2^{-D(\omega)} \leq 2^{-B(m)} < 3^{-m}.$$

Two independent quantities govern the structure of the depth-m cylinder forest: the number $N(m)$ of admissible cylinders and the modulus $2^{D(\omega)}$ associated with each cylinder.

The following section assigns a natural density to each cylinder. Since cylinders at equal depth are pairwise disjoint (Corollary 5.4), the residual density of the depth-m cylinder forest is bounded above by the product of the number of admissible cylinders and the maximal density of any one cylinder.

7 Measure Collapse of the Residual Forest

The admissible forest constructed in the previous sections induces a nested family of subsets of the odd integers. These sets encode the integers whose symbolic trajectories remain admissible to increasing depths.

The central question is whether these residual sets S_m retain positive natural density under repeated symbolic refinement.

7.1 Residual Cylinder Sets

Definition 7.1 (Depth-m Residual Set)**.** Let

$$S_m := \bigcup_{|\omega|=m} C_\omega$$

where the union ranges over all admissible division words of length m.

Remark. The set S_m consists of integers whose valuation trajectories admit an admissible division-word prefix of length m. Equivalently, these are precisely the trajectories that remain unresolved after m refinement steps.

7.2 First-Passage Layers

Definition 7.2 (First-Passage Layer)**.** Define the first-passage layer at depth m by

$$F_m := S_m \setminus S_{m+1}.$$

Remark. The set F_m consists of integers whose symbolic trajectories admit an admissible prefix of length m, but admit no admissible extension to depth $m+1$.

Lemma 7.1 (Disjoint residual decomposition)**.** *For every $m \geq 1$,*

$$S_m = F_m \sqcup S_{m+1},$$

where $\sqcup$ denotes disjoint union.

Proof. By definition,

$$F_m = S_m \setminus S_{m+1}.$$

Hence every element of S_m either:

- remains unresolved to depth $m+1$, or
- exits the residual forest through the first-passage layer F_m

These two possibilities are mutually exclusive, giving the disjoint decomposition. □

Remark (Residual decomposition). The decomposition

$$S_m = F_m \sqcup S_{m+1}$$

shows that symbolic refinement partitions the residual set into two disjoint components: trajectories that exit the residual forest at depth m, and trajectories that remain unresolved to the next level. No trajectories are created or duplicated under refinement.

Corollary 7.2 (Global first-passage decomposition)**.** *The initial residual set admits the disjoint decomposition*

$$S_1 = \left(\bigsqcup_{m \geq 1} F_m \right) \sqcup S_\infty,$$

where

$$S_\infty = \bigcap_{m \geq 1} S_m$$

is the infinite residual set.

Proof. Iterating the decomposition

$$S_m = F_m \sqcup S_{m+1}$$

gives

$$S_1 = F_1 \sqcup F_2 \sqcup \cdots \sqcup F_m \sqcup S_{m+1}.$$

Passing to the limit $m \to \infty$ yields

$$S_1 = \left(\bigsqcup_{m \geq 1} F_m \right) \sqcup S_\infty. \qquad \square$$

Remark (Symbolic conservation law)*.* The global decomposition shows that every residual trajectory either:

- eventually exits into a first-passage layer, or
- survives indefinitely in S_∞.

No symbolic mass is lost or duplicated under refinement; it is partitioned disjointly between finite first-passage escape layers and the infinite residual set.

7.3 Cylinder Measure

Lemma 7.3 (Cylinder Density)**.** *Let C_ω be an admissible cylinder associated with the residue class*

$$r_\omega \pmod{2^{D(\omega)}}.$$

Then

$$\mu(C_\omega) = 2^{-D(\omega)},$$

where μ denotes natural density on the odd positive integers.

Proof. By the Affine Realization Theorem 4.6, C_ω is exactly one residue class modulo $2^{D(\omega)}$. A residue classes modulo $2^{D(\omega)}$ has natural density $2^{-D(\omega)}$. Therefore

$$\mu(C_\omega) = 2^{-D(\omega)}.$$

□

Remark. The density of a cylinder depends only on its modulus $2^{D(\omega)}$, not on the particular residue r_ω. Thus deeper cylinders are exponentially smaller, with each additional power of two halving the natural density.

Lemma 7.4 (Disjointness at fixed depth)**.** *Distinct admissible cylinders of equal depth are disjoint.*

Proof. This follows from the prefix-nesting property established in Section 5. □

Corollary 7.5 (Measure decomposition)**.** *For every m,*

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)}.$$

Remark (Conservation under symbolic refinement)*.* The first-passage decomposition partitions the initial residual set into disjoint escape layers together with the infinite residual set. Thus dyadic mass is neither created nor lost under refinement: it is redistributed between progressively finer symbolic layers.

7.4 Nested residual structure

Definition 7.3 (Infinite Residual Set)**.** Define the infinite residual set by

$$S_\infty := \bigcap_{m=1}^{\infty} S_m.$$

Remark. The set S_∞ consists of integers whose symbolic trajectories admit admissible extensions at every finite depth, equivalently, integers which never exit the residual forest under successive refinement.

Theorem 7.6 (Continuity of measure under nested refinement)**.** *Since*

$$S_{m+1} \subseteq S_m,$$

the measures satisfy

$$\mu(S_\infty) = \lim_{m\to\infty} \mu(S_m).$$

Proof. This follows from continuity from above for natural density on nested measurable residue-class unions. □

7.5 Residual Density Bound

Theorem 7.7 (Residual Density Bound)**.** *For every depth m,*

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)} \leq N(m)\, 2^{-B(m)},$$

where

$$B(m) = A020914(m) = \lfloor m \log_2 3 \rfloor + 1.$$

Proof. By Corollary 7.5,

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)}.$$

By Lemma 5.6,

$$D(\omega) \geq B(m)$$

for every admissible division word of length m. Therefore

$$2^{-D(\omega)} \leq 2^{-B(m)}.$$

Since the sum ranges over $N(m)$ admissible division words of length m,

$$\mu(S_m) \leq N(m)\, 2^{-B(m)},$$

as claimed. $\square$

Theorem 7.8 (Residual Density Collapse)**.** *Suppose the symbolic growth rate satisfies*

$$\lambda < 3,$$

where

$$\lambda = \limsup_{m\to\infty} N(m)^{1/m}.$$

Then

$$\mu(S_m) \longrightarrow 0.$$

Consequently,

$$\mu(S_\infty) = 0.$$

Proof. Let

$$\varepsilon > 0$$

be chosen so that

$$\lambda + \varepsilon < 3.$$

By Lemma 6.1, there exists m_0 such that

$$N(m) \leq (\lambda + \varepsilon)^m$$

for every $m \geq m_0$. Combining this estimate with Theorem 7.7 gives

$$\mu(S_m) \leq (\lambda + \varepsilon)^m\, 2^{-B(m)}.$$

Since

$$2^{B(m)} > 3^m,$$

we have

$$2^{-B(m)} < 3^{-m}.$$

Hence

$$\mu(S_m) < \left(\frac{\lambda + \varepsilon}{3}\right)^m.$$

Because

$$\frac{\lambda + \varepsilon}{3} < 1,$$

the right-hand side converges exponentially to zero. Therefore

$$\mu(S_m) \to 0.$$

Finally, Theorem 7.6 implies

$$\mu(S_\infty) = \lim_{m\to\infty} \mu(S_m) = 0.$$

$\square$

7.6 First-Passage Completeness

Theorem 7.9 (First-Passage Completeness)**.** *Suppose*

$$\lambda < 3.$$

Then

$$\mu(S_1) = \sum_{m=1}^{\infty} \mu(F_m).$$

In particular, if the initial residual set is normalized so that

$$\mu(S_1) = 1,$$

then

$$\sum_{m=1}^{\infty} \mu(F_m) = 1.$$

Thus the set of integers which remain in the residual forest indefinitely has natural density zero.

Proof. By Corollary 7.2,

$$S_1 = \left(\bigsqcup_{m\geq 1} F_m \right) \sqcup S_\infty.$$

Since the union is disjoint,

$$\mu(S_1) = \sum_{m\geq 1} \mu(F_m) + \mu(S_\infty).$$

By Theorem 7.8,

$$\mu(S_\infty) = 0.$$

Therefore

$$\mu(S_1) = \sum_{m \geq 1} \mu(F_m).$$

If the measure is normalized so that $\mu(S_1) = 1$, the stated identity follows immediately. $\square$

7.7 Interpretation

The preceding results reduce the density-one problem for accelerated Collatz trajectories to a purely symbolic question. Every measure-theoretic aspect of the argument is now contained in Theorem 7.8. Consequently, it remains only to establish that the symbolic growth rate of the admissible grammar satisfies

$$\lambda < 3$$

The remainder of the paper is devoted to proving this spectral inequality.

8 The Beatty Barrier

The remaining task is to bound the symbolic growth rate

$$\lambda = \limsup_{m \to \infty} N(m)^{1/m}.$$

The key observation is that the symbolic growth is not free. Every admissible extension of a division word is governed by the Beatty increment sequence associated with

$$B(m) = \lfloor m \log_2 3 \rfloor + 1.$$

Consequently, the admissible branching process is not arbitrary. Every symbolic extension is constrained by the same Beatty sequence that governs the dyadic contraction threshold. Thus the symbolic branching process and the cylinder refinement process are controlled by a common underlying arithmetic.

8.1 Beatty-Controlled Growth

The Beatty increment sequence

$$\delta(m) = B(m+1) - B(m) \in \{1, 2\}$$

governs two parallel constructions:

1. the dyadic refinement of admissible cylinders; and
2. the doubling events that generate the symbolic counts $N(m)$ in the horizontal-sum construction of Appendix E.

The second statement follows from the construction of A186009, whose recursive doubling rule is governed by the increment sequence A022921 [8], equal to the first differences of A020914.

Thus the same Beatty increment sequence governs both admissible cylinder refinement and admissible symbolic branching. The two constructions are therefore synchronized by a common Sturmian grammar.

Because the Beatty increment sequence is Sturmian, it admits canonical return words of lengths $2, 5, 12, \ldots$. Their construction from the continued-fraction expansion of $\log_2 3$ is summarized in Appendix D. For the present section we require only two facts:

1. the canonical return words exist; and
2. each has a well-defined average dyadic increment.

The next subsection shows that, once the increment sequence is fixed, the induced horizontal-sum dynamics is represented by positive linear operators, allowing the local symbolic growth contributed by each return word to be estimated independently.

8.2 Linear Operator Formulation

Let $v_n = (a_1(n), a_2(n), \ldots, a_{k_n}(n))$ denote the generating row at level n, where the length k_n depends on the increment history. Let

$$S_n = \sum_i a_i(n)$$

be its total sum. Away from terminal duplication, the recurrence

$$a_i(n) = a_i(n-1) + a_{i-1}(n)$$

is linear with nonnegative coefficients. The terminal duplication is also linear, so the complete evolution is represented by positive linear operators.

Representing the horizontal-sum dynamics by positive linear operators permits the growth contributed by finite increment blocks to be analyzed through the corresponding operator products.

A single increment acts as a positive linear endomorphism $T_s^{(k)} : \mathbb{R}^k \to \mathbb{R}^k$, preserving row length, whereas a double increment acts as a positive linear map $T_d^{(k)} : \mathbb{R}^k \to \mathbb{R}^{k+1}$, increasing the row length by one via terminal duplication.

The essential feature is positivity: every entry of T_s and T_d is nonnegative. Consequently every admissible evolution is represented by a positive operator product.

Thus each increment symbol $\sigma_n \in \{s, d\}$ induces a positive linear map

$$v_{n+1} = T_{\sigma_n} v_n,$$

so every admissible division word corresponds to a finite product

$$T_{\sigma_{n+k-1}} \cdots T_{\sigma_n}$$

of positive linear operators.

Lemma 8.1 (Controlled Growth of Horizontal Sums)**.** *Let a row have strictly positive entries with total sum S. A single increment preserves the total row sum.*

If a double increment is applied immediately afterward, producing a row in which the terminal entry equals the sum of the previous row and all other entries remain strictly positive, then the total sum of the resulting row satisfies

$$S_{\text{new}} < 3S.$$

Proof. After the intermediate single-increment evolution, the terminal entry equals S, while the remaining interior entries are strictly positive and therefore have total $S_{\text{int}} < S$.

The subsequent double increment duplicates the terminal entry, producing two terminal contributions equal to S. Hence

$$S_{\text{new}} = S_{\text{int}} + 2S,$$

where $S_{\text{int}} < S$. Therefore

$$S_{\text{new}} < 3S.$$

□

Since every admissible evolution is represented by positive operator products, each primitive return block determines a finite operator product whose growth factor depends only on that block. Consequently the symbolic contribution of each primitive return word can be bounded locally.

Appendix C computes these local growth factors for every primitive return block. These provide uniform upper bounds for the symbolic growth contributed by each admissible primitive block.

8.3 Beatty Block Drift

Definition 8.1 (Dyadic Magnitude)**.** For a finite increment block Ω, let

$$\Delta(\Omega)$$

denote the sum of its dyadic increments. Equivalently, $\Delta(\Omega)$ is the total exponent of 2 contributed by the block.

Lemma 8.2 (Beatty Drift Lemma)**.** *Let Ω be a finite block whose average increment satisfies*

$$\frac{\Delta(\Omega)}{|\Omega|} > \log_2 3$$

Then the infinite periodic word

$$\Omega^\infty$$

is not admissible.

Proof. We define Ω^k as the k-fold catentation of Ω. This after k repetitions of Ω,

$$\Delta(\Omega^k) = k\Delta(\Omega)$$

while

$$B(k|\Omega|) = k|\Omega| \log_2 3 + O(1)$$

Since

$$\Delta(\Omega) - |\Omega| \log_2 3 > 0,$$

the difference

$$\Delta(\Omega^k) - B(k|\Omega|)$$

tend to $+\infty$ as $k \to \infty$. Since the excess drift grows linearly while the Beatty discrepancy remains bounded, sufficiently long prefixes violate the admissibility constraints. Therefore the infinite periodic word Ω^∞ is not admissible. $\square$

Remark. By way of an explanation, Ω posesses positive drift relative to $\log_2 3$. Anything with positive drift cannot persist forever because the Beatty sequence never departs from its limiting slope by more than a bounded amount whereas positive drift accumulates linearly without bound.

Lemma 8.3 (Extremal Admissibility Principle)**.** *Let ω be an admissible infinite division word. Then*

$$\limsup_{n\to\infty} \frac{\Delta_\omega(n)}{n} \leq \log_2 3$$

Consequenty no admissible infinite word may contain an asymptotic density of double increments exceeding the Beatty slope.

Proof. Suppose

$$\limsup_{n\to\infty} \frac{\Delta_\omega(n)}{n} > \log_2 3.$$

Then there exists $\varepsilon > 0$ and infinitely many prefixes satisfying

$$\Delta_\omega(n) \geq (\log_2 3 + \varepsilon)n.$$

Hence these prefixes accumulate positive Beatty drift growing linearly with n, contradicting the Beatty Drift Lemma 8.2. □

8.4 Canonical Return Words

The Beatty Drift Lemma 8.2 constrains only the asymptotic average dyadic increment of admissible infinite words. To convert this asymptotic restriction into a finite combinatorial statement, we use the canonical return words of the Beatty Sturmian sequence.

The continued fraction expansion of $\log_2 3$ yield the coefficents, a_k for generating the Sturmian return words according to the recursive equation

$$q_k = a_k q_{k-1} + q_{k-2}$$

For the present argument, only the first two nontrivial canonical return words are required, since every longer canonical return word is assembled recursively from them.

Proposition 8.1 (Recursive Assembly of Canonical Return Words)**.** Let $C_2 = (1,2)$ and $C_5 = (1,2,1,2,2)$ denote the first two nontrivial canonical Sturmian return words. Every subsequent canonical return word is obtained recursively by concatenating copies of C_2 and C_5.

Proof. The canonical Sturmian return words satisfy the standard continued-fraction recursion

$$C_k = C_{k-1}^{a_k} C_{k-2},$$

where a_k is the corresponding continued-fraction coefficient of $\log_2 3$. The next composite word is

$$C_{12} = C_2\, C_5^2$$

Suppose inductively that both C_{k-1} and C_{k-2} are concatenations of C_2 and C_5. Then

$$C_k = C_{k-1}^{a_k} C_{k-2}$$

is likewise a concatenation of the same primitive words. The result follows by induction. □

8.5 The Beatty Barrier Theorem

The Beatty increment sequence satisfies

$$a(n) = \lfloor (n+1)\log_2(3) \rfloor - \lfloor n \log_2 3 \rfloor$$

and therefore

$$\lim_{m\to\infty} \frac{1}{m} \sum_{k=1}^{m} a(k) = \log_2 3$$

as recorded in A020914. Among the primitive return words, the block

$$C_5 = (1,2,1,2,2)$$

has an avergage increment

$$\frac{8}{5} > \log_2 3.$$

Hence the Beatty Drift Lemma implies that the infinite periodic word

$$(1,2,1,2,2)^{\infty}$$

is not admissible. The obstruction is purely arithmetic: the excess average increment accumulates linearly, while the Beatty discrepancy remains bounded.

Theorem 8.4 (Beatty Barrier Theorem)**.** *No admissible infinite division word can contain the primitive block*

$$C_5 = (1, 2, 1, 2, 2)$$

with asymptotic frequency one. Equivalently, every admissible infinite word must contain infinitely many longer return blocks whose average dyadic increment is strictly smaller than $8/5$*.*

Proof. Suppose the asymptotic frequency is one. Then

$$\frac{\Delta(n)}{n} \to \frac{8}{5}$$

since

$$\frac{8}{5} > \log_2 3.$$

this contradicts the Extremal Admissibility Principle (Lemma 8.3), which asserts that every admissible infinite division word satisfies

$$\limsup_{n\to\infty} \frac{\Delta_\omega(n)}{n} \le \log_2 3$$

Therefore C_5 cannot occur with asymptotic frequency one. $\square$

9 Spectral Growth Bound

The admissible grammar is constrained by the Beatty growth law $B(n)$. Any symbolic process whose average increment exceeds the Beatty slope must eventually violate admissibility. The periodic 5-cycle is the minimal non-trivial expansive process with this property.

Appendix C computes the comparison growth contributed by every primitive return block. Among these,

$$C_5 = (1, 2, 1, 2, 2)$$

has the largest per-symbol comparison growth. Nevertheless,

$$\Lambda_5 = \left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5} < 3.$$

Lemma 9.1 (Extremal Block Dilution). *Every occurrence of an expansive primitive contributes positive Beatty drift. Since total drift must remain bounded, expansive primitives must be balanced by sufficiently many contractive return words. Consequently their asymptotic frequency is bounded away from one.*

Proof. By the Extremal Admissibility Principle (Lemma 8.3), every admissible infinite division word has average dyadic increment at most $\log_2 3$.

The primitive block C_5 has average increment $8/5 > \log_2 3$, so each occurrence contributes positive Beatty drift relative to the admissible mean.

If the asymptotic frequency of expansive blocks approached one, the average drift would exceed the Beatty slope, contradicting the Extremal Admissibility Principle.

Hence expansive primitives must be balanced by return blocks having smaller average increment. Consequently the asymptotic frequency of expansive blocks is bounded away from one. □

Theorem 9.2 (Spectral Growth Bound). *Every admissible symbolic trajectory satisfies*

$$\lambda < 3.$$

Proof. The Beatty Barrier Theorem excludes infinite repetition of the unique extremal primitive block.

Every admissible trajectory therefore contains infinitely many return blocks with strictly smaller average growth. By the Spectral Compensation Lemma 9.1 any instance of the 5-block one must insert at least one compensating block.

Since every compensating block has comparison growth strictly smaller than the extremal 5-block, the asymptotic exponential growth rate is strictly below the growth obtained by perpetual repetition of C_5. Then

$$\Lambda_5 < 3.$$

Thus every admissible symbolic trajectory satisfies

$$\lambda < 3.$$

□

9.1 Interpretation

The Sturmian analysis identifies the primitive return words that may appear in the Beatty grammar. The proof of the spectral bound depends only on the Beatty drift restriction established in the previous section. The continued-fraction and return-word analysis serves only to justify that the return cycles are canonical primitive comparison words rather than ad hoc choices.

The Beatty-controlled affine grammar admits an extremal local growth mechanism whose infinite repetition is structurally forbidden, and that forbidden mechanism is already subcritical, namely $\lambda < 3$.

Thus the measure of the residual forest collapses as $n \to \infty$.

10 Conclusion

This paper reformulates the accelerated Collatz problem in terms of an admissible symbolic grammar whose arithmetic realization is given by nested dyadic cylinders. The resulting cylinder forest provides an exact symbolic representation of the odd integers together with a natural measure-theoretic filtration.

Within this framework the measure of the residual forest is controlled by two competing exponential processes: symbolic proliferation and dyadic contraction. The measure-theoretic analysis shows that these combine through a single spectral parameter λ. Consequently, proving

$$\lambda < 3$$

is sufficient to force the residual forest to have natural density zero.

Combining the spectral estimate with the residual density bounds shows that the infinite residual set has natural density zero. Equivalently, the first-passage layers exhaust the entire normalized dyadic measure.

The principal contribution is therefore not merely a new estimate, but a reformulation of the density problem itself. Rather than studying individual Collatz trajectories directly, the argument transfers the problem through three equivalent descriptions:

$$\text{symbolic grammar} \longrightarrow \text{cylinder geometry} \longrightarrow \text{measure dynamics}.$$

Within this framework, the density question becomes a problem of controlling the spectral growth of an admissible Sturmian grammar.

This synthesis of symbolic dynamics, Diophantine approximation, and dyadic measure provides a unified framework in which the density question for the accelerated Collatz map reduces to a single explicit spectral inequality.

A The Canonical Affine Map

Each division word ω induces an affine map $F_\omega(x)$ of length m (see Section 2)

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}, \qquad D(\omega) = d_0 + \cdots + d_{m-1},$$

where $c(\omega)$ is the affine constant canonically determined by the division word ω. The associated congruence constraint is naturally considered modulo $2^{D(\omega)}$.

Although the existence and uniqueness of $c(\omega)$ follows from the affine representation developed in Section 2, the explicit computation is useful for concrete examples and software implementations. The following procedure reconstructs the same affine constant $c(\omega)$ from any representative of the associated residue class.

A.1 Canonical Choice

For a fixed ω, any integer x in the corresponding residue class satisfies

$$3^m x + c \equiv 0 \pmod{2^{D(\omega)}}.$$

We remove ambiguity by selecting the unique representative

$$0 \leq c(\omega) < 2^{D(\omega)}.$$

This choice depends only on ω and is independent of the representative x.

A.2 Step-by-Step Computation

Given ω of length m and total division count $D(\omega)$:

1. Choose any representative x of the residue class determined by ω.
2. Compute $y = 3^m x$.
3. Let $r = \lceil y/2^{D(\omega)} \rceil$ be the smallest integer such that $r \cdot 2^{D(\omega)} \geq y$.
4. Define $c(\omega) = r \cdot 2^{D(\omega)} - y$.

By this construction, $0 \leq c(\omega) < 2^{D(\omega)}$ ensuring that

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}} \in \mathbb{Z} \quad \text{for all } x \text{ in the class.}$$

A.3 Example

Let $\omega = (1, 1, 1, 4)$ with $m = 4$ and $D(\omega) = \sum_{i=0}^{3} d_i = 1 + 1 + 1 + 4 = 7$:

1. Choose $x = 15$. Compute $y = 3^4 \cdot 15 = 1215$.
2. $2^7 = 128$, so $r = \lceil 1215/128 \rceil = 10$.
3. $c(\omega) = 10 \cdot 128 - 1215 = 65$.

Hence the canonical map is

$$F_\omega(x) = \frac{81x + 65}{128}.$$

Checking another representative $x = 143$:

$$F_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91,$$

confirming that the same $c(\omega)$ works for the entire residue class. The admissibility condition becomes

$$81x + 65 \equiv 0 \pmod{128}.$$

Solving yields the unique residue class

$$x \equiv 15 \pmod{128}.$$

Hence the cylinder is

$$x(k) = 15 + 128k.$$

Applying the affine map,

$$F_\omega(15 + 128k) = 81k + 10.$$

Thus the symbolic cylinder generated by ω is mapped onto the arithmetic progression

$$10,\ 91,\ 172,\ 253, \ldots$$

with common difference $81 = 3^4$.

Remark

This construction is purely illustrative and is not required for the density or growth arguments in the main text. It demonstrates concretely how the affine map is fully determined by the division word ω.

By the Cylinder Representation Theorem (Section 5), every realization of ω belongs to a single residue class modulo $2^{D(\omega)}$. Consequently, the value of $c(\omega)$ computed above is independent of the chosen representative.

Remark (Cylinder Rigidity). The division word ω uniquely determines its canonical affine constant $c(\omega)$, and therefore uniquely determines the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

Since admissible realizations of ω occupy a single residue class modulo $2^{D(\omega)}$, the symbolic data encoded by ω simultaneously determines both the affine dynamics and the associated cylinder.

A.4 Illustrative Derivation of the Affine Constant

For readers wishing to see the affine constant arise directly from iterated substitution, consider a general division word

$$\omega = (a, b, c, d)$$

of length 4.

The accelerated relations are

$$x_1 = \frac{3x_0 + 1}{2^a}, \qquad x_2 = \frac{3x_1 + 1}{2^b}, \qquad x_3 = \frac{3x_2 + 1}{2^c}, \qquad x_4 = \frac{3x_3 + 1}{2^d}.$$

Eliminating intermediate variables recursively yields

$$2^{a+b+c+d} x_4 = 3^4 x_0 + 3^3 + 3^2 2^a + 3\, 2^{a+b} + 2^{a+b+c}.$$

Hence

$$F_\omega(x) = \frac{3^4 x + \left(3^3 + 3^2 2^a + 3\, 2^{a+b} + 2^{a+b+c}\right)}{2^{a+b+c+d}},$$

so that

$$c(\omega) = 3^3 + 3^2 2^a + 3\, 2^{a+b} + 2^{a+b+c}.$$

This illustrates concretely how the affine constant accumulates the propagated contributions of successive "+1" terms throughout the accelerated iteration.

B Continued Fractions, Tower Cycles, and the Extremal Envelope

This appendix records the Diophantine structure underlying the admissible grammar and explains the origin of the distinguished cycle lengths

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335,\dots$$

that appear throughout the paper.

B.1 Continued-Fraction Origin of the Tower Cycles

Let

$$\alpha = \log_2 3.$$

Its continued-fraction expansion begins

$$\alpha = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, \dots].$$

Writing

$$\frac{p_k}{q_k}$$

for the convergents of α, the first denominator sequence is

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\dots$$

which coincides exactly with the tower lengths observed in the admissible grammar. The convergent denominators satisfy the standard recurrence

$$q_k = a_k q_{k-1} + q_{k-2},$$

where a_k is the corresponding continued-fraction coefficient. Consequently every tower cycle is assembled canonically from the two preceding tower cycles.

Table 1 records the first instances.

Table 1: Continued-fraction generation of the tower cycles.

q_k	Cycle	k	a_k	Cycle Composition	Sequence
1	C_0	0	–	Single	1
1	C_1	1	1	Double	2
2	C_2	2	1	$C_0 + 1 \cdot C_1$	$1, 2$
5	C_5	3	2	$2 \cdot C_2 + C_1$	$1, 2, 1, 2, 2$
12	C_{12}	4	2	$C_2 + 2 \cdot C_5$	$1, 2, 1, 2, 1, 2, 2, 1, 2, 1, 2, 2$
41	C_{41}	5	3	$3 \cdot C_{12} + C_5$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
53	C_{53}	6	1	$C_{12} + 1 \cdot C_{41}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
306	C_{306}	7	5	$5 \cdot C_{53} + C_{41}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
665	C_{665}	8	2	$C_{53} + 2 \cdot C_{306}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
15601	C_{15601}	9	23	$23 \cdot C_{665} + C_{306}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
31867	C_{31867}	10	2	$C_{665} + 2 \cdot C_{15601}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
79335	C_{79335}	11	2	$2 \cdot C_{31867} + C_{15601}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$

Thus the tower hierarchy is not an empirical feature of the grammar. It is forced by the Diophantine approximation structure of $\log_2 3$.

B.2 Alternating Expansion and Contraction

The convergents alternately overestimate and underestimate α.

Define

$$\varepsilon_k = q_k \alpha - p_k.$$

Then

$$\varepsilon_k \to 0$$

and the signs alternate.

Since

$$\alpha = \log_2 3,$$

it follows that

$$\varepsilon_k = q_k \log_2 3 - p_k = -\log_2 \left(\frac{2^{p_k}}{3^{q_k}} \right).$$

Equivalently,

$$\frac{2^{p_k}}{3^{q_k}} = 2^{-\varepsilon_k}.$$

Hence

$$\left|\log\left(\frac{2^{p_k}}{3^{q_k}}\right)\right| = (\log 2)\,|\varepsilon_k| \to 0.$$

Therefore

$$\frac{2^{p_k}}{3^{q_k}} \to 1.$$

The convergents alternately overestimate and underestimate $\log_2 3$, so the ratios alternately lie above and below unity while approaching exact multiplicative balance.

Table 2: Convergents of $\log_2 3$ and resulting dyadic–ternary synchronization ratios.

q_k	p_k	$\varepsilon_k = q_k \log_2(3) - p_k$	$2^{p_k}/3^{q_k}$
1	2	$-0.41594\ldots$	$1.33333\ldots$
2	3	$+0.16993\ldots$	$0.88889\ldots$
5	8	$-0.07519\ldots$	$1.05350\ldots$
12	19	$+0.01955\ldots$	$0.98654\ldots$
41	65	$-0.01654\ldots$	$1.01153\ldots$
53	84	$+0.00301\ldots$	$0.99791\ldots$

B.3 Primitive Return Words and the Failure of the Double Process

The Sturmian word generated by

$$d(n) = \lfloor n\alpha \rfloor + 1$$

admits a return-word decomposition. At the first nontrivial level, the primitive return words have lengths 2, 5, and 12. These correspond respectively to:

$$C_2 = (1, 2).$$

$$C_5 = (1, 2, 1, 2, 2).$$

$$C_{12} = (1, 2, 1, 2, 1, 2, 2, 1, 2, 1, 2, 2).$$

The 2-cycle is locally contractive, while the 5-cycle is locally expansive. A 12-cycle is formed by concatenating the 2-cycle with a pair of consecutive 5-cycles, but the 12-cycle is overall contractive.

One might ask why the extremal comparison process is based on the 5-cycle rather than the shorter expansive 1-cycle corresponding to a double increment.

Indeed, the shortest expansive block is

$$C_1 = (2),$$

for which

$$\frac{2^2}{3} = \frac{4}{3} > 1.$$

Repeated doubles generate the row sums

$$1, 2, 5, 14, 42, 132, 429, \ldots$$

which are the Catalan numbers. Since

$$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}}$$

implies

$$\lim_{n\to\infty} \frac{C_{n+1}}{C_n} = 4.$$

Thus an infinite concatenation of doubles produces exponential growth rate 4, which exceeds the critical value 3.

The pure-double process is therefore unsuitable as a comparison envelope: it overestimates growth by too large a margin and cannot yield the desired spectral bound.

B.4 The Extremal Envelope

The main text proves that every admissible Sturmian word decomposes uniquely into primitive return words. Since the normalized cocycle contribution of C_5 exceeds that of every other primitive return word, replacing each return word by C_5 can only increase the average cocycle drift. Although the periodic comparison word C_5^∞ is itself forbidden by the Sturmian grammar, it provides a universal comparison envelope. The explicit evaluation of this envelope is carried out in Appendix C.

C Extremal Local Growth Calculations

This appendix provides the explicit finite computations underlying the local growth bounds used in the density argument. Recall that the sequence A020914 has first differences given by A022921, consisting of repeated blocks of *single* steps (s) and *double* steps (d).

C.1 Illustrative Extremal Computations

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n), \quad i, n \in \mathbb{N}, \tag{C.1}$$

so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ only in that the terminal entry of the row is duplicated, causing the final term to appear twice in the generating tuple while preserving monotonicity of the preceding entries.

Lemma C.1 (Pascal-Type Horizontal Growth)**.** *Let a row generated by the horizontal-sum construction have strictly positive entries with total sum S.*

After one horizontal-sum step without duplication, the resulting row has total

$$S_{\text{new}} = S + S_{\text{int}},$$

where S_{int} denotes the sum of the nonterminal entries of the previous row.

Consequently

$$S_{\text{new}} < 2S.$$

Proof. The terminal entry of the new row equals the total sum S of the previous row. Every nonterminal entry contributes exactly once to the new row through the horizontal-sum recurrence. Hence

$$S_{\text{new}} = S + S_{\text{int}}.$$

Since all entries are strictly positive,

$$S_{\text{int}} < S,$$

and therefore

$$S_{\text{new}} < 2S.$$

$\square$

Corollary C.2 (Controlled Growth Under Duplication)**.** *A subsequent duplication contributes at most one additional copy of the terminal entry, whose value is S.*

Hence

$$S_{\text{new}} < 2S + S = 3S.$$

These examples are chosen so that each step saturates the inequalities of Lemma C.1, thereby realizing extremal growth. Equalities shown below should be interpreted as extremal upper-bound realizations rather than identities valid for all rows.

The following row-by-row computations realize extremal growth for the admissible local words and together determine the sharpest uniform bounds used in Proposition D.1. Each example illustrates how a local pattern can saturate its maximal growth factor under the horizontal-sum dynamics. Let S denote the total sum of entries in the generating row prior to a given local pattern.

Example 1.

$$\textit{Single } a(10) = \underbrace{173}_{\text{sum:=}S} : \underbrace{1, 7, 25, 55, 85}_{S}$$

$$\textit{Double } a(11) = \underbrace{476}_{\text{sum=}3S} : \underbrace{1, 8, 33, 88}_{S}\underbrace{173}_{S}, \underbrace{173}_{S}$$

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum=}9S} : \underbrace{1, 9, 42, 130}_{2S}\underbrace{303}_{3S}, \underbrace{476}_{4S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum=}33S} : \underbrace{1, 10, 52, 182}_{4S}, \underbrace{485}_{7S}, \underbrace{961}_{11S}, \underbrace{961}_{11S}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum}=123S} : \underbrace{1, 11, 63, 245}_{8S}, \underbrace{730}_{15S}, \underbrace{1691}_{26S}, \underbrace{2652}_{37S}, \underbrace{2652}_{37S}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum}=329S} : \underbrace{1, 12, 75, 320}_{16S}, \underbrace{1050}_{31S}, \underbrace{2741}_{57S}, \underbrace{5393}_{94S}, \underbrace{8045}_{131S}$$

The resulting maximal local ratios are

Pattern	Max Ratio
s, d	$3/1$
d, d	$123/33$
s, d, s	$9/3$
d, d, s	$329/123$

Example 2.

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum}:=S} : \underbrace{1, 9, 42, 130, 303, 476}_{S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum}=3S} : \underbrace{1, 10, 52, 182, 485}_{S}, \underbrace{961}_{S}, \underbrace{961}_{S}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum}=13S} : \underbrace{1, 11, 63, 245, 730}_{2S}, \underbrace{1691}_{3S}, \underbrace{2652}_{4S}, \underbrace{2652}_{4S}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum}=37S} : \underbrace{1, 12, 75, 320, 1050}_{4S}, \underbrace{2741}_{7S}, \underbrace{5393}_{11S}, \underbrace{8045}_{15S}$$

$$\textit{Double } a(16) = \underbrace{51033}_{\text{sum}=131S} : \underbrace{1, 13, 88, 408, 1458}_{8S}, \underbrace{4199}_{15S}, \underbrace{9592}_{26S}, \underbrace{17637}_{41S}, \underbrace{17637}_{41S}$$

$$\textit{Single } a(17) = \underbrace{108950}_{\text{sum}=341S} : \underbrace{1, 14, 102, 510, 1968}_{16S}, \underbrace{6167}_{31S}, \underbrace{15759}_{57S}, \underbrace{33396}_{98S}, \underbrace{51033}_{139S}$$

The corresponding maximal local ratios are

Pattern	Max Ratio
s, d	$3/1$
d, d	$13/3$
s, d, s	$341/131$
d, d, s	$37/13$

Improved Extremal Alignment

Because the total multiplicative growth over a cycle is the product of the local growth factors, the geometric mean over one full cycle is maximized by selecting the maximal admissible factor at each local pattern occurrence. Consequently, once a uniform upper bound has been established for each admissible local pattern, the extremal global growth rate over a complete cycle is obtained by multiplying these maximal local factors according to their frequencies and taking the corresponding geometric mean.

Equivalently, because total growth over any admissible word is multiplicative in its local factors, the asymptotic per-symbol growth rate is the geometric mean of these factors; hence the maximal rate over the entire language generated by the grammar is achieved by the word that maximizes this mean subject to the grammar constraints.

Each example yields a valid (not necessarily sharp) upper bound on the maximal growth factor associated with each admissible local pattern. Since these bounds arise from the same uniform recurrence and apply globally to all admissible configurations, the true maximal growth factor for a given pattern is bounded above by the smallest recorded bound among all such valid upper bounds.

Because the recurrence is monotone with respect to the placement of delayed doubling, no other admissible configuration can exceed these extremal realizations. Accordingly, combining the sharpest bounds obtained across different initiating rows yields the tightest global envelope on admissible local growth.

Pattern	Max Ratio	Associated coefficient count
s, d	3	2
d, d	123/33	1
s, d, s	341/131	1
d, d, s	329/123	1

Consequently, the maximal geometric growth rate of a 5–cycle satisfies

$$\Big(3^2 \cdot (123/33) \cdot (341/131) \cdot (329/123)\Big)^{1/5} \approx 2.976 < 3. \tag{C.2}$$

Conclusion. Every admissible growth pattern of A186009 is constrained by the grammar of A022921. Infinite concatenations of 5-cycles, though forbidden, saturate local growth constraints and provide a strict upper envelope. Equation (C.2) confirms that even this extremal envelope has geometric growth rate < 3, ensuring subdominance relative to the denominator $2^{B(i)}$.

D Extremal Growth in the Sturmian Return-Word System

D.1 Dyadic Threshold Sequence Sturmian Structure

Theorem D.1 (Sturmian realization of the threshold grammar)**.** *Let*

$$B(n) = \lfloor n \log_2 3 \rfloor + 1.$$

Then the increment sequence

$$\delta(n) = B(n+1) - B(n)$$

is a Sturmian word over the alphabet $\{1, 2\}$*. Consequently the admissible grammar inherits the combinatorial structure of an irrational rotation.*

Proof. This is the classical characterization of Beatty increment sequences associated with irrational slopes. □

Let

$$\alpha := \log_2 3, \qquad d(n) := \lfloor n\alpha \rfloor + 1.$$

The increment sequence

$$\delta(n) := d(n+1) - d(n) \in \{1, 2\}$$

is a Sturmian word of slope $\alpha - 1$ and intercept 0. Equivalently, δ defines a symbolic dynamical system

$$X_\alpha \subset \{1, 2\}^{\mathbb{N}}$$

generated by an irrational rotation. All admissible growth structures considered in this paper are functions of this Sturmian coding.

D.2 Return Words and Primitive Cycle Decomposition

Definition D.1 (Return Words)**.** A finite word C is a return word of the Sturmian system X_α if it is the minimal block between successive visits to a fixed cylinder set induced by δ. We denote by $\mathcal{R}_\alpha$ the set of return words.

Theorem D.2 (Return Words via Continued Fractions)**.** *Let $\alpha = \log_2 3$ and let $\delta(n)$ be the Sturmian coding induced by rotation R_α.*

Let q_k denote the denominators of the continued-fraction convergents of α. For the prefix defining the extremal admissible growth process, the set of return words is exactly:

$$\mathcal{R}_\alpha = \{C_{q_k}, C_{q_{k+1}}\}$$

for some index k determined by the chosen prefix. In particular, for the admissible threshold structure of $d(n) = \lfloor n\alpha \rfloor + 1$, the relevant convergent pair is:

$$(q_k, q_{k+1}) = (5, 12).$$

Proof. The sequence $\delta(n)$ is a Sturmian word of slope α. Return words to any prefix correspond to first-return maps of the rotation R_α to a subinterval I.

The convergent denominators begin (see Appendix B for details):

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335, \ldots$$

For the prefix relevant to the admissible grammar, the associated return times are the consecutive denominators 5 and 12. Hence the return-word set is $\{C_5, C_{12}\}$. □

Although the shortest return word is the single double increment $C_1 = (2)$, repeated doubles generate the Catalan sequence $1, 2, 5, 14, 42, 132, \ldots$ whose asymptotic exponential growth rate is 4. Since this exceeds the critical threshold 3, the pure-double process cannot serve as a useful comparison envelope.

Proposition D.1 (Extremal Primitive Comparison Word)**.** The return word C_5 is the shortest return word whose induced comparison growth is strictly less than 3.

Remark (Diophantine Origin of the Grammar)*.* The admissible grammar is governed not by arbitrary combinatorial rules but by the Diophantine approximation properties of $\log_2 3$. The convergent denominators

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335, \ldots$$

arise from the continued-fraction approximants of

$$\alpha = \log_2 3,$$

and consequently, the primitive return words governing admissible refinement are determined by best rational approximations to the Beatty slope rather than by ad hoc combinatorial choices.

$$\frac{p_k}{q_k} \approx \log_2 3.$$

Equivalently,

$$2^{p_k} \approx 3^{q_k},$$

so the associated multiplicative cocycle satisfies

$$\frac{2^{p_k}}{3^{q_k}} \to 1.$$

In particular, the appearance of the 5-, 12-, 41-, and 53-cycles is not accidental; these cycles are induced by the Diophantine approximation hierarchy of $\log_2 3$.

Lemma D.3 (Induced Return-Word Hierarchy)**.** *Every admissible word in the Sturmian language generated by*

$$\delta(n) = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$$

admits a decomposition into return words generated recursively from the primitive pair (C_5, C_{12})*.*

D.3 Multiplicative Cocycle Structure

Define the multiplicative growth cocycle

$$G(w) := \frac{2^{\#s(w)+2\#d(w)}}{3^{|w|}},$$

where $\#s(w)$ and $\#d(w)$ are the number of single and double increments in w.

Lemma D.4 (Cocycle Additivity)**.** *For concatenation of admissible return words,*

$$G(C_{i_1} \cdots C_{i_n}) = \prod_{k=1}^{n} G(C_{i_k}).$$

Equivalently,

$$\log G(w) = \sum_k \log G(C_{i_k}).$$

Thus the global exponential growth rate reduces to weighted averages of return-word contributions.

D.4 Local Evaluation of Return Words

The local comparison constants computed in Appendix C yield the following per-symbol cocycle contributions. The two primitive return words satisfy:

$$\frac{1}{5}\log G(C_5) = \log\left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5},$$

$$\frac{1}{12}\log G(C_{12}) < \frac{1}{5}\log G(C_5).$$

Hence C_5 is the unique extremal return word with maximal per-symbol drift.

Proposition D.2 (Extremal Cocycle Principle)**.** For any admissible infinite word $\delta \in X_\alpha$,

$$\limsup_{n\to\infty} \frac{1}{n}\log G(\delta_{[1,n]}) \le \frac{1}{5}\log G(C_5).$$

Proof. By the unique decomposition into return words and cocycle additivity, the growth rate is a convex combination of the per-symbol contributions of C_5 and C_{12}.

Since

$$\frac{1}{5}\log G(C_5) > \frac{1}{12}\log G(C_{12}),$$

the maximal comparison average is obtained by replacing every return word with C_5, corresponding formally to the periodic comparison word C_5^∞. □

D.5 Global Growth Bound

Theorem D.5 (Spectral Growth Bound)**.** *Let λ denote the symbolic growth rate introduced in Section 6.*

Then

$$\lambda \le \exp\left(\frac{1}{5}\log G(C_5)\right) < 3.$$

Proof. By Proposition D.1, every admissible infinite word has average cocycle growth bounded above by that of C_5^∞. Thus

$$\lambda \leq \exp\left(\frac{1}{5}\log G(C_5)\right).$$

Via explicit evaluation of the local cocycle contributions (Appendix C),

$$\left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5} < 3.$$

$\square$

Theorem D.5 closes the logical chain developed throughout the paper. The Beatty threshold sequence induces the admissible grammar; the grammar determines the symbolic growth rate λ; the Sturmian return-word analysis establishes the strict bound $\lambda < 3$; and Theorem 7.8 then forces the residual forest to have zero natural density. Thus the symbolic, arithmetic, and measure-theoretic components of the argument fit together in a single chain of implications.

E Horizontal-Sum Table for A186009

For reference, a portion of A186009 generated by the horizontal-sum construction is given below as per an algorithm derived from Winkler [8]. This table is included for illustration of the horizontal-sum mechanism; none of the density or growth bounds in the main text depend on these specific numeric values.

Table 3: A186009 Horizontal Sum Formulation

n	$a_1(n)$	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$	$a_7(n)$	$a_8(n)$	$a_9(n)$	$\sum a_i(n)$
1	1									1
2	1									1
3	1									1
4	1	1								2
5	1	2								3
6	1	3	3							7
7	1	4	7							12
8	1	5	12	12						30
9	1	6	18	30	30					85
10	1	7	25	55	85					173
11	1	8	33	88	173	173				476
12	1	9	42	130	303	476				961
13	1	10	52	182	485	961	961			2652
14	1	11	63	245	730	1691	2652	2652		8045
15	1	12	75	320	1050	2741	5393	8045		17637
16	1	13	88	408	1458	4199	9592	17637	17637	51033
17	1	14	102	510	1968	6167	15759	33396	51033	108950

There is a sequence closely related to A186009 called A100982. A186009 prepends the number 1 to the A100982 sequence. Along with the listing of initial terms for A100982 comes an algorithm which explains how to generate the entire sequence starting with an initial value of 1. As a result of its close similarity, this method can also be leveraged to generate all of the terms in A186009.

Theorem 2 [8] referenced in A100982 asserts that the elements in the sequence can be derived in a Pascal–triangle–type manner, beginning with an initial condition:

$$a(1) = a_1(1) = 1$$

The algorithm for $A100982(n)$ states that the k^{th} term of the n^{th} series component can be computed as follows:

$$a_{k+1}(n) = a_k(n) + a_k(n-1); k \in \mathbb{N}_0, n \in \mathbb{N}$$

The algorithm also stipulates that the last term in the sequence is repeated whenever $A022921(n-2) = 2$ (which can be derived from A020914 by taking the first differences between its terms). In doing so, this imposes the implicit constraint that this rule applies only when $n \geq 2$. Keeping in mind that repeated terms in A022921 occur only once $n \geq 2$, the recurrence is extended to $n = 1$ by treating the duplication condition as inactive at the boundary, producing an augmented vertical-sum expansion.

In rows without terminal duplication, this recurrence is exactly Pascal's binomial rule; duplication acts only at the boundary and therefore preserves the monotone, binomial-type growth of interior entries. This boundary-only modification is precisely why the extremal growth analysis in Appendix C reduces to *local pattern frequencies* rather than global row shape.

We reformulate this construction such that the first non-zero term in each row is always at index 1 and thus the formula henceforth (producing the same expansion components up to a reindexing of coordinates) is modified as follows.

Preserving $a(1) = a_1(1) = a_1(n) = 1$ for all $n \in \mathbb{N}$, the i^{th} term of the n^{th} A100982 element is:

$$a_i(n) = a_i(n-1) + a_{i-1}(n); i, n \in \mathbb{N}$$

This reindexing does not change any row sums or terminal-duplication positions; it only relocates coordinates. Using this formulation the initial term for all values of $n \in \mathbb{N}$ is always 1 rather than shifting out for each value of n. The other significant difference is that in this formulation the values for $a_i(n)$ appear horizontally as opposed to vertically. The doubling rule for the terminal entry remains unchanged in substance – namely that it occurs when $A022921(n-2) = 2$ - which is precisely the single/double grammar governing admissible growth in the preceding appendices, and hence the local multiplicative structure used in the extremal bounds.

Because A186009 is obtained from A100982 by prepending an initial row, all row indices shift by one. Consequently the terminal-duplication condition expressed in the original formulation as $A022921(n-2) = 2$ now becomes $A022921(n-3) = 2$ in the present indexing.

References

[1] Lothar Collatz. On the iteration of numerical functions. *Mathematische Zeitschrift*, 1937. In German.

[2] Jeffrey C. Lagarias. *The Ultimate Challenge: The 3x+1 Problem*, volume 55 of *Student Mathematical Library*. American Mathematical Society, 2010.

[3] Richard Terras. A stopping time problem on the positive integers. *Acta Arithmetica*, 30:241–252, 1976.

[4] Tomás Oliveira e Silva. Maximum excursion and stopping time record-holders for the 3x+1 problem. *Mathematics of Computation*, 68(225):371–384, 1999.

[5] Terence Tao. Almost all collatz orbits attain almost bounded values. *Forum of Mathematics, Pi*, 8:e14, 2020. doi: 10.1017/fmp.2020.7.

[6] Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics*. Addison-Wesley, 2 edition, 1994.

[7] M. Lothaire. *Algebraic Combinatorics on Words*, volume 90 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2002.

[8] Mike Winkler. The algorithmic structure of the finite stopping time behavior of the $3x + 1$ function. *arXiv e-prints*, arXiv:1709.03385, November 2017. Version 3.